

Exp No. 3: Spectrum Analysis of Sub-1 GHz, Mid-band & mmWave by Plotting Propagation

AIM: To analyze the propagation loss characteristics of different frequency bands (Sub-1 GHz, Midband, and mmWave) in wireless communication systems and compare their performance over varying distances using MATLAB.

Objective 1: To compute the Propagation Loss (dB) for Sub-1 GHz, Mid-band, and mmWave frequency ranges using standard path loss models.

Program

```
clc;

clear;

close all;

distance = 0.1:0.1:5;

frequency = [900 3500 28000];

loss900 = 32.44 + 20*log10(distance) + 20*log10(frequency(1));

loss3500 = 32.44 + 20*log10(distance) + 20*log10(frequency(2));

loss28000 = 32.44 + 20*log10(distance) + 20*log10(frequency(3));

figure

subplot(2,1,1)

plot(distance,loss900,'b','LineWidth',2)

hold on

plot(distance,loss3500,'r','LineWidth',2)

plot(distance,loss28000,'k','LineWidth',2)

title('Propagation Loss vs Distance')

xlabel('Distance (km)')

ylabel('Loss (dB)')

grid on

subplot(2,1,2)

freqAxis = frequency;

lossValue = [loss900(10) loss3500(10) loss28000(10)];
```

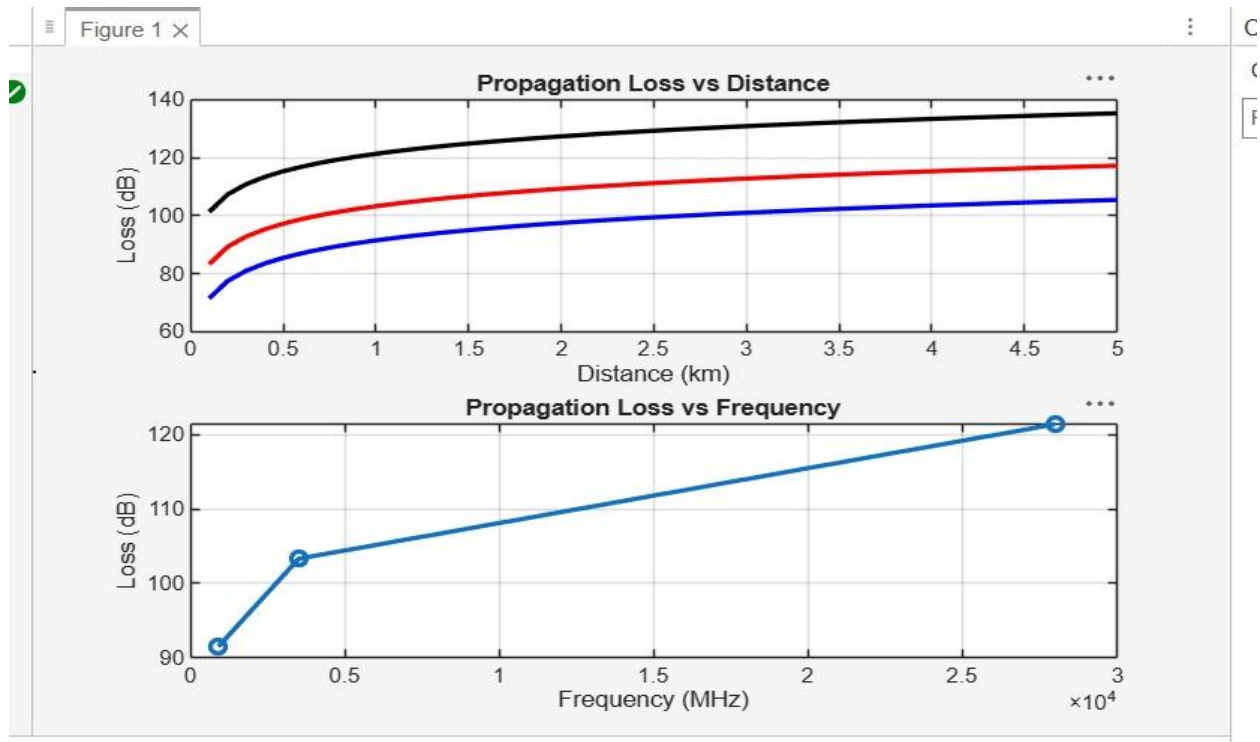
```
plot(freqAxis,lossValue,'-o','LineWidth',2)
```

```
title('Propagation Loss vs Frequency')
```

```
xlabel('Frequency (MHz)')
```

```
ylabel('Loss (dB)')
```

```
grid on
```



Program

```
clc;
```

```
clear;
```

```
close all;
```

```
distance = 2;
```

```
freq = [900 3500 28000];
```

```
pathLoss = 32.44 + 20*log10(distance) + 20*log10(freq);
```

```
figure
```

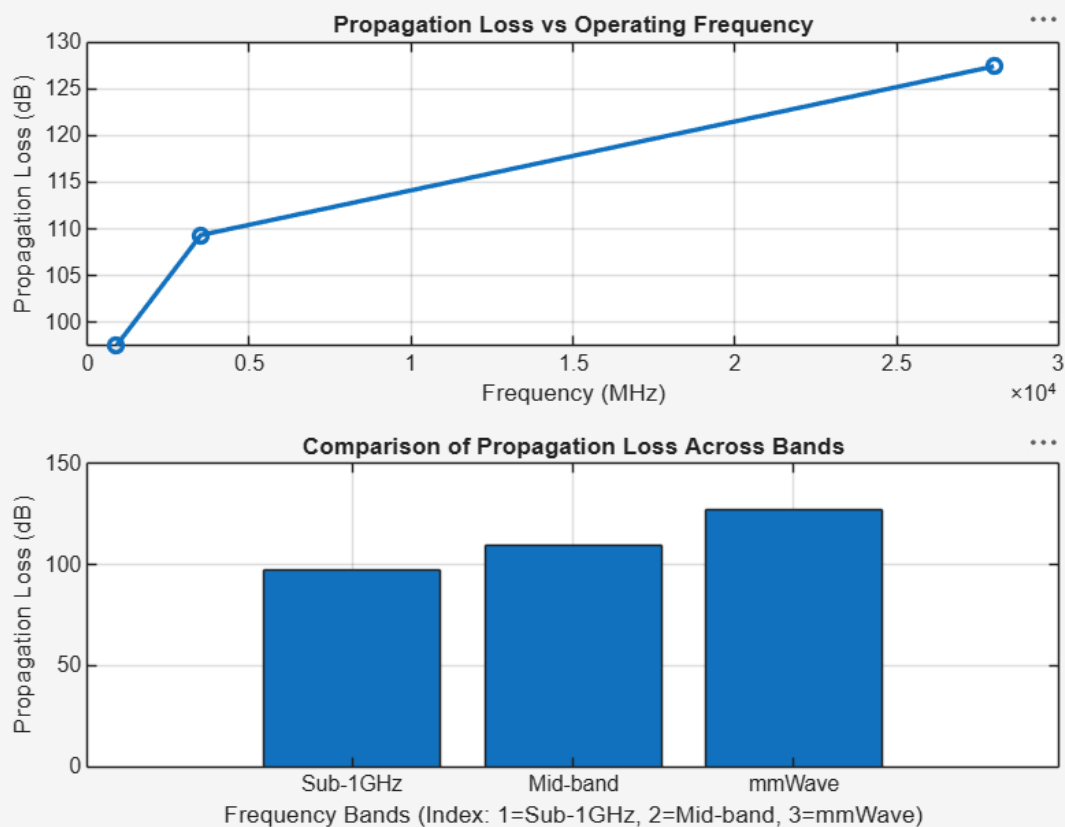
```
subplot(2,1,1)
```

```
plot(freq,pathLoss,'-o','LineWidth',2)
```

```

title('Propagation Loss vs Operating Frequency')
xlabel('Frequency (MHz)')
ylabel('Propagation Loss (dB)')
grid on
subplot(2,1,2)
lossData = pathLoss;
bar(lossData)
title('Comparison of Propagation Loss Across Bands')
xlabel('Frequency Bands (Index: 1=Sub-1GHz, 2=Mid-band, 3=mmWave)')
ylabel('Propagation Loss (dB)')
set(gca,'XTickLabel',{'Sub-1GHz','Mid-band','mmWave'})
grid on

```



Objective 3: To analyze how **Communication Distance (km)** affects propagation loss across different spectrum bands.

program

```
clc;

clear;

close all;

freqValue = [900 3500 28000];

linkDistance = 2;

loss = 32.44 + 20*log10(linkDistance) + 20*log10(freqValue);

figure

subplot(2,1,1)

plot(freqValue,loss,'-o','LineWidth',2)

title('Propagation Loss vs Operating Frequency')

xlabel('Frequency (MHz)')

ylabel('Propagation Loss (dB)')

grid on

subplot(2,1,2)

band = 1:length(freqValue);

plot(band,loss,'-s','LineWidth',2)

title('Band-wise Propagation Loss Comparison')

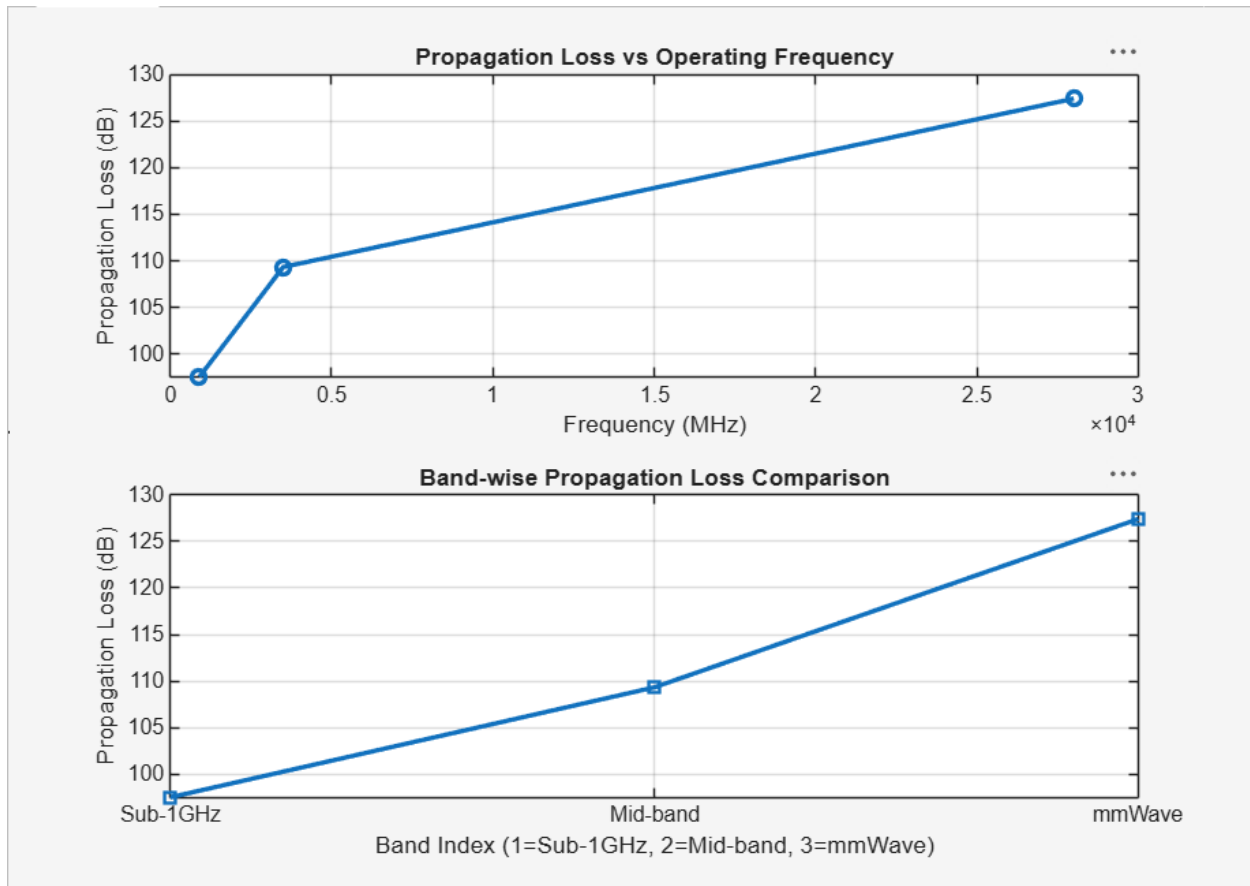
xlabel('Band Index (1=Sub-1GHz, 2=Mid-band, 3=mmWave)')

ylabel('Propagation Loss (dB)')

set(gca,'XTick',band)

set(gca,'XTickLabel',{'Sub-1GHz','Mid-band','mmWave'})

grid on
```



Result:

- 1.The propagation loss was successfully computed for Sub-1 GHz, Mid-band, and mmWave frequency bands using the Free Space Path Loss (FSPL) model.
- 2.It was observed that propagation loss increases with increase in operating frequency, with mmWave showing the highest attenuation and Sub-1 GHz showing the lowest loss.
- 3.The comparative analysis of spectrum bands clearly demonstrates that lower frequency bands are more suitable for long-range communication, while higher frequency bands require advanced techniques like beamforming for efficient performance.